Test Case / Original Request

Run PTC-004 using K6. Execute the real performance test and collect monitoring results. Assess the security surface for the same job and exact configured target.Analyze the current job’s results and generate the final PDF with security coverage and evidence, plus links to the native JMeter report, raw results, and final PDF.

REAL Execution Outcome

Execution status: COMPLETED
Runner status: completed
Outcome details: REAL k6 execution completed. Threshold breaches, if any, are retained as performance findings.
Thresholds passed: True

Executive Summary

Application: Unknown
Scenario: Filtered Feedback API — Load Check
Job ID: JOB-992dd506a393453c8bf507c83bbb449d
Overall Result: WARNING

Performance was adequate in terms of throughput and zero errors, but there was significant latency variance: p95 latency was much higher than average and outlier spikes were noted, triggering warnings. Infrastructure health during the run was stable with low CPU usage and high—but not alarming—memory usage, and no monitoring anomalies or alerts were detected. Security checks raised a warning due to a missing Strict-Transport-Security header, reflecting only a single observed response and not a persistent control gap. No historical context was available, so analysis is based solely on evidence from this run.

Execution Profile and Artifacts

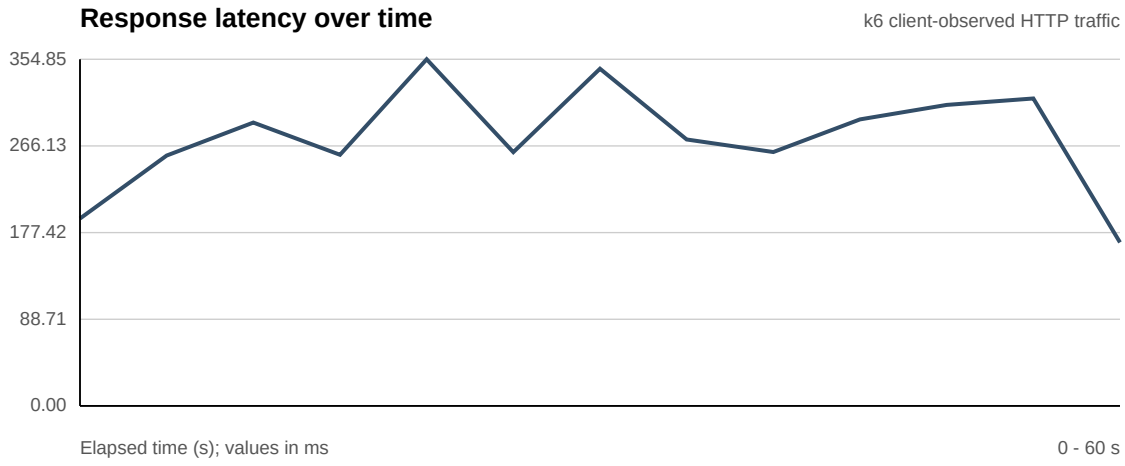
Execution mode: REAL
Workload profile:
- tool: k6
- target_url: https://jsonplaceholder.typicode.com
- endpoint: /comments?postId=1
- method: GET
- users: 20
- max_rps: 12
- duration_seconds: 60
- ramp_up_seconds: 5
- expected_status_code: 200
- p95_threshold_ms: 5000
Artifact directory: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d
Artifact references:
- aligned_time_series: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_aligned_time_series.csv
- generated_script: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check.js
- k6_summary: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_summary.json

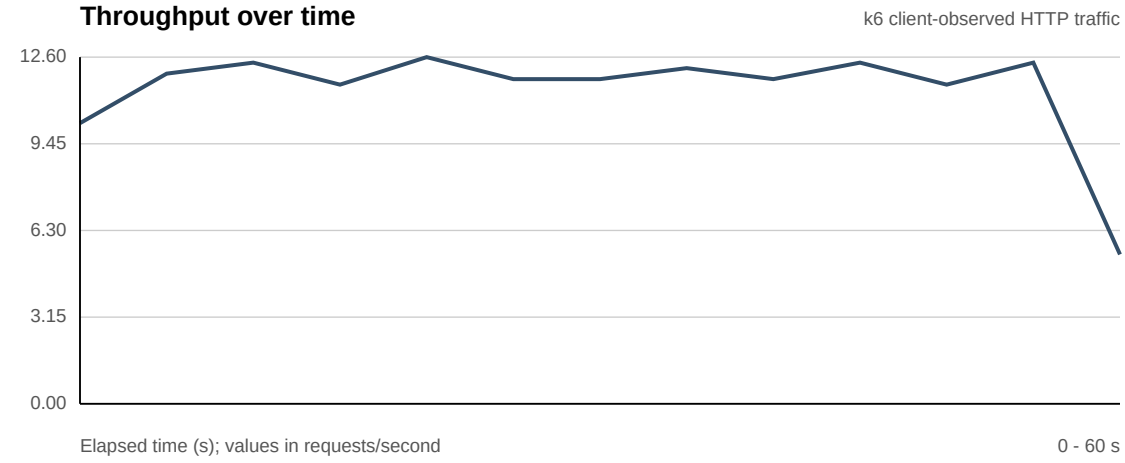
- measurement_metadata: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_measurement_metadata.json
- monitoring_samples: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_runtime_samples.csv
- Native k6 HTML report
- Native report + raw results ZIP
- normalized_metrics: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_normalized_metrics.json
- Raw k6 results (JSON)
- runner_log: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_runner.log
- time_series: src/generated_scripts/k6 native reports/JOB-992dd506a393453c8bf507c83bbb449d/k6_PTC_004_Filtered_Feedback_API_Load_Check_time_series.csv

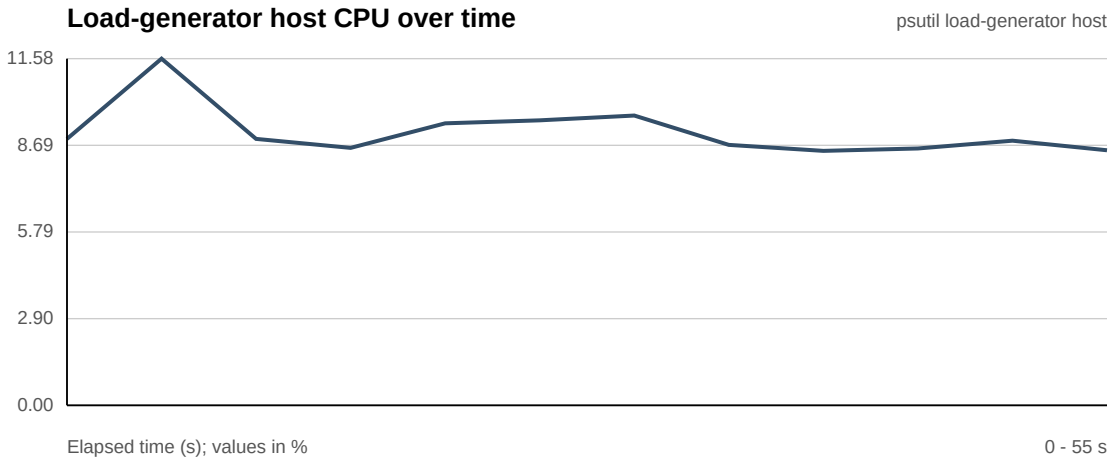
The interactive native HTML dashboard is linked as a separate artifact and is not embedded in this PDF.

Time-Series Charts (REAL Measurements)

Charts show measured values in five-second aligned buckets. Host CPU is from the load-generator machine, not the target.







Agent Activity Summary

Agent	*Summary*
Test Execution	Load test executed on /comments?postId=1 (Unknown - Filtered Feedback API — Load Check). Results: 715 total requests, 715 successful, 0 failed. Avg response time: 285.04082657342644 ms; 95th percentile: 1030.899799999999 ms; Max: 5796.213 ms. Throughput: 11.89116566552476 TPS; Error rate: 0.0%.
Monitoring	CPU usage: 9.15% average, 17.1% peak; memory: 77.9% average, 78.8% peak; DB connections: not collected, TPS: 11.89116566552476, Response time: 285.04082657342644 ms. Data source and scope: REAL: psutil load-generator host samples and k6 client-observed metrics. Measurement provenance: cpu_usage=psutil (load_generator_host); cpu_peak_usage=psutil (load_generator_host); memory_usage=psutil (load_generator_host); memory_peak_usage=psutil (load_generator_host); response_time=k6 (load_generator_observed_http_traffic); tps=k6 (load_generator_observed_http_traffic); db_connections=not_collected (No target-specific database telemetry provider is configured.). No alerts triggered.

Agent	*Summary*
Security Validation	Status: WARNING. Live security check: completed. Issue: Strict-Transport-Security was not present in the observed response; this single response does not establish the full origin policy. Issue: Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness. Issue: Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness. Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0} Control assessment evidence: - external_surface: assessed; basis=live_observation; evidence=['live-observation:JOB-992dd506a393453c8bf507c83bbb449d'] Evidence references: live-observation:JOB-992dd506a393453c8bf507c83bbb449d Recommendation: Review whether an HSTS policy is appropriate for this HTTPS API origin.
Analysis	Performance was adequate in terms of throughput and zero errors, but there was significant latency variance: p95 latency was much higher than average and outlier spikes were noted, triggering warnings. Infrastructure health during the run was stable with low CPU usage and high—but not alarming—memory usage, and no monitoring anomalies or alerts were detected. Security checks raised a warning due to a missing Strict-Transport-Security header, reflecting only a single observed response and not a persistent control gap. No historical context was available, so analysis is based solely on evidence from this run.
Reporting	A PDF report was generated, including all findings, recommendations, and an executive summary.

Per-API Findings (Test Execution)

- /comments?postId=1 [WARNING / high]
avg=285.04082657342644ms p95=1030.899799999999ms max=5796.213ms throughput=11.89116566552476 req/s error_rate=0.0% http_status=200
* p95 (1030.899799999999ms) is 3.6x the average (285.04082657342644ms) -- wide latency variance / possible tail contention.
* Max response time (5796.213ms) is 5.6x p95 (1030.899799999999ms) -- outlier spikes present.
history: Historical context unavailable due to backend lookup issue, so only static thresholds were applied. Latency variance and outlier spikes are considered anomalous based on current run data alone.
recommendation: Investigate backend or external factors causing tail latency and response outliers; since no relevant SLA or incident context could be loaded, focus on current threshold breaches and observed behavior.

Infrastructure Health (Monitoring)

Status: PASS (monitoring_agent reported: OK)
CPU usage: 9.15% Memory usage: 77.9% DB connections: None TPS: 11.89116566552476 Response time (monitoring): 285.04082657342644ms

Security Findings (Security Agent)

Status: WARNING Security score: N/A
Security result job ID: JOB-992dd506a393453c8bf507c83bbb449d.
Live security check: completed.
Issues:
- Strict-Transport-Security was not present in the observed response; this single response does not establish the full origin policy.
- Scope limitation: configuration validator outcomes describe profile declarations; any live result is a separate single-request observation and does not prove control effectiveness.
- Live HEAD observation completed with HTTP status 200; response headers are observations only, not proof of control effectiveness.
Control coverage: {'requested_count': 1, 'assessed_count': 1, 'not_assessed_count': 0}
Requested controls: external_surface
Assessed controls: external_surface
Control evidence and provenance:
- external_surface: assessed; basis=live_observation; evidence fields=['target_origin', 'endpoint', 'method', 'response_status_code', 'observed_security_headers']; references=['live-observation:JOB-992dd506a393453c8bf507c83bbb449d']
Evidence references: live-observation:JOB-992dd506a393453c8bf507c83bbb449d
Live observation: HEAD /comments?postId=1; HTTP 200; elapsed 488.9099169522524 ms; requests sent 1.
Observed response headers (observations only):
- cache-control: max-age=43200
- x-content-type-options: nosniff
Recommendations:
- Review whether an HSTS policy is appropriate for this HTTPS API origin.

Recommended Next Steps

- /comments?postId=1: Investigate backend or external factors causing tail latency and response outliers; since no relevant SLA or incident context could be loaded, focus on current threshold breaches and observed behavior.
- Security: address the issues/recommendations listed above before next deployment.